



Advanced Time Series Analysis and Forecasting

Seminar 5: Bayesian VAR, factor models and nowcasting



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BBE: Bernanke–Boivin–Eliasz (FAVAR, 2005)
- ▣ BGR: Bańbura–Giannone–Reichlin (large Bayesian VARs, 2010)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVAR: Bayesian Vector AutoRegression
- ▣ CPI: Consumer Price Index
- ▣ DFM: Dynamic Factor Model
- ▣ EM: Expectation–Maximisation (algorithm)
- ▣ ESI: Economic Sentiment Indicator (European Commission)
- ▣ ESS: Effective Sample Size
- ▣ FAVAR: Factor-Augmented Vector AutoRegression
- ▣ FFR: Federal Funds Rate
- ▣ FRED: Federal Reserve Economic Data
- ▣ FRED-MD: FRED Monthly Database for macroeconomic research (McCracken–Ng)
- ▣ GDP: Gross Domestic Product
- ▣ GLP: Giannone–Lenza–Primiceri (hierarchical prior selection, 2015)
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ IC: Interval de încredere (confidence interval)
- ▣ IP: Industrial Production
- ▣ MCMC: Markov Chain Monte Carlo
- ▣ MEDIUM: the 20-variable system of BGR
- ▣ MIDAS: MIXed DATA Sampling (regression)
- ▣ NIW: Normal–inverse–Wishart (prior)
- ▣ OLS: Ordinary Least Squares
- ▣ PCA: Principal Component Analysis

Acronyms used in this material (2/2)

- ▣ RMSE: Root Mean Squared Error
- ▣ ROBOR: Romanian Interbank Offer Rate
- ▣ RW: Random Walk
- ▣ SMALL: the 3-variable system of BGR (employment, CPI, federal funds rate)
- ▣ US: United States
- ▣ VAR: Vector AutoRegression

Today's question and route

- **Question:** how do we estimate, forecast and nowcast with many series and few observations?
 - ▶ this seminar comes **before** Lecture 5: the section “Prerequisites for Today” gives every definition the tasks use
- **Route**
 - ▶ Part A: conjugate updating, the Minnesota prior, Gibbs diagnostics, dummy observations, principal components, Bai–Ng, the Mariano–Murasawa weights, news
 - ▶ Part B: hyperparameters by marginal likelihood; a Romanian BVAR; factors on three samples; FAVAR and the number of factors; nowcasting Romanian GDP; news
 - ▶ Part C: do surveys still help after 2020?; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	conjugate updating and empirical Bayes; Minnesota prior variances and a ridge posterior	Solved, Proposed	A1
A3, A4	Gibbs conditionals and MCMC diagnostics; dummy observations	Solved, Proposed	A1, A3
A5, A6	principal components of an equicorrelated panel; Bai–Ng criteria	Solved, Proposed	A5
A7, A8	Mariano–Murasawa weights; news and MIDAS weights	Solved, Proposed	A7
B1, B2	hyperparameters by marginal likelihood (US); a Romanian BVAR	Solved, Proposed	B1
B3, B4	factors on three samples; FAVAR and the number of factors	Solved, Proposed	B3
B5, B6	nowcasting Romanian GDP; the news of September 2026	Solved, Proposed	B5
C1, C2	surveys after 2020; what is wrong in an AI answer?	Proposed	B5, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
FRED-MD-format US panel (117 series)	FRED	monthly	1960–2026
RO IP, HICP, ROBOR 3M, unemployment, ESI	Eurostat	monthly	2005–2026
EUR/RON	BNR reference rate	monthly average	2005–2026
RO nowcasting panel (11 series), real GDP	Eurostat	monthly, quarterly	2003–2026

- In the notebook: `bvar_fit`, `glp_optimize`, `niw_logml`, `em_pca`, `bai_ng`, `favar_run`, `ro_vintage`, `bridge_nowcast`, `midas_nowcast`, `dfm_nowcast`, `news_decomposition`; no account or key is needed

Prerequisites for Today (1/7): Bayesian updating

- Bayes: $p(\theta | y) \propto p(y | \theta)p(\theta)$: posterior $\propto$ likelihood $\times$ prior
 - ▶ θ : the parameters; y : the data; marginal likelihood $p(y) = \int p(y | \theta)p(\theta) d\theta$: the density of the data averaged over the prior
- Normal prior $\beta \sim N(m_0, v_0)$, data $\hat{\beta} | \beta \sim N(\beta, s^2)$: the posterior is $\beta | y \sim N(m_1, v_1)$
 - ▶ $\hat{\beta}$: the OLS estimate, with standard error s ; m_0, v_0 : prior mean and variance
 - ▶ precisions (1/variance) add: $v_1^{-1} = v_0^{-1} + s^{-2}$; means average: $m_1 = v_1(m_0/v_0 + \hat{\beta}/s^2)$
 - ▶ marginal likelihood: $\hat{\beta} \sim N(m_0, v_0 + s^2)$ once β is integrated out

Prerequisites for Today (2/7): Gibbs sampling and MCMC diagnostics

- ▣ **Gibbs:** draw each block from its full conditional (its distribution given the other blocks and the data), and repeat
 - ▶ regression $y = X\beta + e$: $\beta \mid \sigma^2 \sim N$, $\sigma^2 \mid \beta \sim IG$ (inverse gamma)
- ▣ Effective sample size $ESS = M / (1 + 2 \sum_k \rho_k)$
 - ▶ M : number of draws; ρ_k : autocorrelation of the draws at lag k ; ESS: the number of independent draws worth the same
- ▣ $\hat{R} = \sqrt{\hat{V}/W}$, $\hat{V} = \frac{n-1}{n} W + \frac{B}{n}$ Gelman and Rubin (1992)
 - ▶ several chains of n draws; W : the within-chain variance; $B = n \times$ the variance of the chain means
 - ▶ $\hat{R}$ close to 1 (below 1.01) and a large ESS are necessary, not sufficient, for convergence

Prerequisites for Today (3/7): the Minnesota prior

- VAR(p) as a regression $Y = XB + U$
 - ▶ Y ($T \times n$): the data; X : constant and p lags; B : coefficients; U : errors with covariance Σ ; $(A_l)_{ij}$: effect of variable j at lag l on variable i
- Minnesota prior [Litterman \(1986\)](#): means δ_i on the own first lag, 0 elsewhere; variances λ^2/l^2 (own lags), $\lambda^2\sigma_i^2/(l^2\sigma_j^2)$ (other variables)
 - ▶ $\delta_i = 1$: random walk; λ : overall tightness (small λ : strong shrinkage); σ_i^2 : residual variance of an AR for y_i (fixes the units)
- Natural conjugate form [Kadiyala and Karlsson \(1997\)](#): $\text{vec}B \mid \Sigma \sim N(\text{vec}B_0, \Sigma \otimes \Omega_0)$, $\Sigma \sim IW(\Psi, d)$
 - ▶ B_0 : prior mean; Ω_0 : prior covariance across regressors; $\otimes$: Kronecker product; IW : inverse Wishart
 - ▶ posterior mean $\bar{B} = (X'X + \Omega_0^{-1})^{-1}(X'Y + \Omega_0^{-1}B_0)$: a precision-weighted average of OLS and B_0

Prerequisites for Today (4/7): dummy observations and hyperparameters

- ▣ **Dummy observations** Bańbura, Giannone and Reichlin (2010)
 - ▶ (Y_d, X_d) : artificial rows that state the prior as if it were data
 - ▶ OLS on $\begin{pmatrix} Y_d \\ Y \end{pmatrix}, \begin{pmatrix} X_d \\ X \end{pmatrix}$ = the posterior mean with $\Omega_0 = (X_d' X_d)^{-1}$, $B_0 = \Omega_0 X_d' Y_d$
- ▣ Sum of coefficients Doan, Litterman and Sims (1984): $Y_d = \text{diag}(\bar{y})/\mu$, $X_d = (1_p' \otimes \text{diag}(\bar{y})/\mu, 0)$
 - ▶ $\bar{y}$: the means of the first p observations; 1_p : a vector of ones; a small μ pushes towards unit roots
 - ▶ initial observation Sims and Zha (1998): one row with $\bar{y}/\phi$; a small ϕ pushes towards unit roots or a stationary mean $\bar{y}$
- ▣ **Hyperparameters** (λ, μ, ϕ) by the marginal likelihood $p(Y \mid \lambda, \mu, \phi)$ with Gamma hyperpriors Giannone, Lenza and Primiceri (2015)

Prerequisites for Today (5/7): factor models

- Factor model $x_{it} = \lambda_i' F_t + e_{it}$: each series = common part + idiosyncratic part
 - ▶ x_{it} : series i at time t (N series, T periods); F_t : the r factors; λ_i : the loadings of series i ; e_{it} : the idiosyncratic error
- PCA: $\hat{F} = \sqrt{T} \times$ first r eigenvectors of XX' (standardised data), $\hat{\Lambda} = X'\hat{F}/T$ [Stock and Watson \(2002a\)](#)
 - ▶ X ($T \times N$): the panel; only the factor space is identified: $(\Lambda H, H^{-1}F)$ fits equally well for any invertible H
- **Bai–Ng** [Bai and Ng \(2002\)](#): $IC_{p2}(k) = \ln V(k) + k \frac{N+T}{NT} \ln \min(N, T)$
 - ▶ $V(k)$: residual mean square after k components; the second term penalises each extra factor; choose k minimising the criterion
- **FAVAR** [Bernanke, Boivin and Elias \(2005\)](#)
 - ▶ a VAR in $(\hat{F}_t, Y_t)$; Y_t : the policy rate; the slow factors are purged of Y_t
 - ▶ response of any series x_i = its loadings $\times$ the factor responses

Prerequisites for Today (6/7): mixed frequencies

- ▣ Quarterly growth from monthly growth [Mariano and Murasawa \(2003\)](#):

$$y_t^Q \approx \frac{1}{3}(y_t + 2y_{t-1} + 3y_{t-2} + 2y_{t-3} + y_{t-4})$$

- ▶ y_t : monthly growth of a latent monthly GDP; y_t^Q : quarter-on-quarter growth, observed in the third month of the quarter
- ▶ bridge: complete the quarter with AR forecasts, aggregate, regress [Baffigi, Golinelli and Parigi \(2004\)](#)

- ▣ **MIDAS** [Ghysels, Santa-Clara and Valkanov \(2006\)](#): $y_\tau = \beta_0 + \beta_1 \sum_j w_j(\theta) x_{m_\tau-j} + \varepsilon_\tau$, $w_j \propto e^{\theta_1 j + \theta_2 j^2}$

- ▶ τ : the quarter; $x_{m_\tau-j}$: the monthly indicator j months before the latest month m_τ ; $w_j(\theta)$: weights summing to 1, shaped by $\theta = (\theta_1, \theta_2)$
- ▶ one regression per information set (position in the quarter)

Prerequisites for Today (7/7): the DFM and news

- ▣ **DFM** (dynamic factor model)
 - ▶ $x_t = \Lambda f_t + e_t$, and the factors f_t follow a VAR
 - ▶ the Kalman filter handles missing data and the ragged edge (series ending in different months)
- ▣ **News** Bańbura and Modugno (2014): $\text{revision} = \sum_j w_j l_j$, $w = \mathbb{E}[y l'] \mathbb{E}[l l']^{-1}$
 - ▶ Ω_{old} : the information before the new releases x_j ; $l_j = x_j - \mathbb{E}[x_j \mid \Omega_{old}]$: the news, i.e. the unexpected part; l : their vector
 - ▶ w_j : the weight of release j in the revision of the nowcast of y ; an expected release ($l_j = 0$) changes nothing

A1: conjugate updating and empirical Bayes [Solved]

Task

- ▣ AR(1) without constant, $\sigma^2 = 1$ known, $\sum x_t^2 = 50$, $\sum x_t y_t = 40$; prior $\rho \sim N(1, 0.2^2)$.
- ▣ 1. Compute $\hat{\rho}$ and its standard error s .
- ▣ 2. Compute the posterior mean, standard deviation and the weight of the prior.
- ▣ 3. With the prior $\rho \sim N(0, \tau^2)$, find the τ^2 that maximises the marginal likelihood of $\hat{\rho}$.
- ▣ Report: two numbers, three numbers, one number.

Solution

- ▣ 1. $\hat{\rho} = 40/50 = 0.800$, $s = 1/\sqrt{50} = 0.141$
- ▣ 2. Precisions $25 + 50 = 75$: $m_1 = (25 \cdot 1 + 50 \cdot 0.8)/75 = 0.8667$, $sd = 1/\sqrt{75} = 0.1155$, prior weight $25/75 = 0.333$
- ▣ 3. $\hat{\rho} \sim N(0, \tau^2 + s^2)$; maximising in τ^2 : $\tau^2 + s^2 = \hat{\rho}^2$, so $\hat{\tau}^2 = \max(0, \hat{\rho}^2 - s^2) = 0.64 - 0.02 = 0.620$
- ▣ Interpretation: the data choose how much to shrink; this is the one-parameter version of GLP

A2: Minnesota prior variances and a ridge posterior [Proposed]

Solution discussed in class

Task

- ▣ Bivariate VAR(2), equation 1; $\lambda = 0.2$, $\sigma_1 = 1$, $\sigma_2 = 2$, $\delta_1 = 1$. Model: A1.
- ▣ 1. Give the prior standard deviations of the own and cross coefficients at lags 1 and 2.
- ▣ 2. Keep only $y_{1,t-1}$ and $y_{2,t-1}$: $X'X = \begin{pmatrix} 100 & 20 \\ 20 & 400 \end{pmatrix}$, $X'y = (85, 30)'$, $\sigma^2 = 1$; compute the OLS and the posterior means.
- ▣ 3. Say what happens as $\lambda \rightarrow 0$ and as $\lambda \rightarrow \infty$.
- ▣ Report: four numbers, two pairs of numbers, two sentences.



A3: Gibbs conditionals and MCMC diagnostics [Solved]

Task

- ▣ $y = X\beta + e$, $e \sim N(0, \sigma^2 I)$, priors $\beta \sim N(b_0, V_0)$, $\sigma^2 \sim IG(a_0, d_0)$, independent.
- ▣ 1. Derive $\beta \mid \sigma^2, y$ and $\sigma^2 \mid \beta, y$.
- ▣ 2. A chain of $M = 10\,000$ draws has AR(1) autocorrelation $\rho = 0.9$: compute the ESS.
- ▣ 3. Four chains of $n = 1\,000$ draws have means $(0.10, 0.12, 0.35, 0.11)$ and within variance $W = 0.01$: compute $\hat{R}$.
- ▣ Report: two densities, one number, one number and a verdict.

Solution

- ▣ 1. $\beta \mid \cdot \sim N(\bar{b}, \bar{V})$, $\bar{V} = (V_0^{-1} + X'X/\sigma^2)^{-1}$,
 $\bar{b} = \bar{V}(V_0^{-1}b_0 + X'y/\sigma^2)$; $\sigma^2 \mid \cdot \sim IG(a_0 + T/2, d_0 + e'e/2)$
- ▣ 2. $\sum_k \rho^k = \rho/(1 - \rho)$: ESS = $M(1 - \rho)/(1 + \rho) = 526.3$ (one effective draw every 19)
- ▣ 3. $B = n \text{var}(\bar{\theta}_c) = 14.467$, $\hat{V} = 0.999W + B/n = 0.0245$,
 $\hat{R} = 1.56$: not converged; chain 3 is stuck elsewhere (without it,
 $\hat{R} = 1.004$)

A4: dummy observations [Proposed]

Solution discussed in class

Task

- The AR(1) of A1; Minnesota dummy $y_d = \sigma\delta/\lambda$, $x_d = \sigma/\lambda$ with $\sigma = 1$, $\delta = 1$, $\lambda = 0.2$. Model: A1, A3.
- 1. Show that OLS on the augmented data equals $(X'X + X_d'X_d)^{-1}(X'y + X_d'y_d)$.
- 2. Compute it and compare with A1.
- 3. Add a sum-of-coefficients dummy $y = x = \bar{y}/\mu$ with $\bar{y} = 5$, $\mu = 1$ and recompute; say where it pushes ρ .
- Report: one derivation, one number, one number and one sentence.



A5: principal components of an equicorrelated panel [Solved]

Task

- ▣ $N = 3$ standardised series with all pairwise correlations $\rho = 0.6$.
- ▣ 1. Find the eigenvalues and eigenvectors of the correlation matrix R .
- ▣ 2. Compute the share of variance of the first component and its loadings.
- ▣ 3. Give the share for $N = 100$ and its limit as $N \rightarrow \infty$.
- ▣ Report: three eigenvalues, one share, one share and a limit.

Solution

- ▣ 1. $R = (1 - \rho)I + \rho\mathbf{1}\mathbf{1}'$: $\lambda_1 = 1 + 2\rho = 2.2$ with $1/\sqrt{3}$; $\lambda_{2,3} = 1 - \rho = 0.4$
- ▣ 2. Share $2.2/3 = 73.3\%$; loadings $1/\sqrt{3} = 0.577$ each: the factor is the cross-sectional average
- ▣ 3. $(1 + 99\rho)/100 = 60.4\%$, limit ρ : the first eigenvalue grows like N , the others stay bounded (pervasiveness)

A6: the number of factors by Bai and Ng [Proposed]

Solution discussed in class

Task

- ▣ $N = 100$, $T = 200$; $V(k)$ for $k = 0, \dots, 5$: 1.000, 0.620, 0.480, 0.430, 0.415, 0.405. Model: A5.
- ▣ 1. Compute the IC_{p2} penalty per factor.
- ▣ 2. Compute $IC_{p2}(k)$ for each k and choose k .
- ▣ 3. Explain why $V(k)$ alone always picks $k = 5$, and why the chosen factors are defined only up to a rotation.
- ▣ Report: one number, a table of six values and k , two sentences.

A7: the Mariano–Murasawa weights [Solved]

Task

- ▣ Quarterly GDP is the 3-month average of a latent monthly level Y_t^M ; $y_t = \Delta \ln Y_t^M$.
- ▣ 1. Approximate $\ln Y_t^Q$ by the average of the three monthly logs.
- ▣ 2. Derive $y_t^Q = \ln Y_t^Q - \ln Y_{t-3}^Q$ as a combination of $y_t, \dots, y_{t-4}$.
- ▣ 3. With $(y_t, \dots, y_{t-4}) = (0.4, 0.3, -0.1, 0.5, 0.2)$, compute y_t^Q .
- ▣ Report: one formula, five weights, one number.

Solution

- ▣ 1. Geometric mean instead of the arithmetic mean:
 $\ln Y_t^Q \approx \frac{1}{3} \sum_{j=0}^2 \ln Y_{t-j}^M$
- ▣ 2. Write each log difference over three months as a sum of monthly growth rates: weights $(1, 2, 3, 2, 1)/3$, summing to 3
- ▣ 3. $(0.4 + 0.6 - 0.3 + 1.0 + 0.2)/3 = 0.633$

A8: news and MIDAS weights [Proposed]

Solution discussed in class

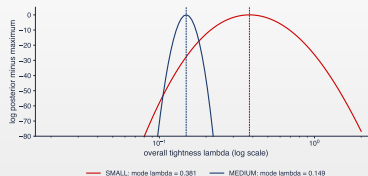
Task

- ▣ Standardised y (GDP growth) and survey release x , $\text{Cov}(y, x) = 0.6$, $\text{Var}x = 1$; $\mathbb{E}[x \mid \Omega_{old}] = 0.2$, released $x = -0.8$. Model: A7.
- ▣ 1. Compute the news and the revision of the nowcast.
- ▣ 2. Two releases with $\mathbb{E}[ll'] = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$, $\mathbb{E}[y/l] = (0.6, 0.3)'$: compute the weights and say why the second is zero.
- ▣ 3. Exponential Almon with $\theta = (0.5, -0.2)$, $K = 4$: compute the weights.
- ▣ Report: two numbers, two weights and a sentence, four weights.

B1: hyperparameters by marginal likelihood [Solved]

- ▣ **Question:** how tight should the Minnesota prior be for the US SMALL and MEDIUM systems?
- ▣ **Inputs:** SMALL (private employment, CPI, funds rate) and MEDIUM (20 series of BGR), $100 \times \log$ levels or levels, $p = 13$, 1960:1–2019:12
- ▣ **Tasks**
 1. Compute the posterior mode of (λ, μ, ϕ) with the GLP Gamma hyperpriors for each system.
 2. Plot the log posterior as a function of λ and report its loss at $\lambda = 0.2$ and $\lambda = 1$.
 3. Compute the sum of the own-lag coefficients of CPI in the posterior mean (and by OLS for SMALL).
 4. Interpretation: what do the modes say about the size of the system and about unit roots?
- ▣ **Report:** six hyperparameters, four losses, three sums, one sentence
- ▣ **Notebook:** section B1

B1: solution [Solved]



System	λ	μ	ϕ	loss at 0.2 / 1	CPI own sum
SMALL ($k = 40$)	0.381	0.199	0.440	-12.6 / -26.3	0.9999 (OLS 0.9980)
MEDIUM ($k = 261$)	0.149	0.134	0.445	-42.6 / -2173.5	1.0000

- Interpretation: the larger system wants a tighter prior
 - ▶ λ 0.381 against 0.149; the BGR fit rule gives 0.158
 - ▶ small μ and ϕ : the data favour persistence and near unit roots
 - ▶ the CPI own-lag sum is almost exactly 1

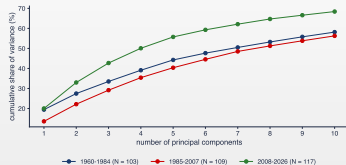
B2: a Bayesian VAR for Romania [Proposed]

- **Question:** does a GLP-type BVAR forecast Romanian inflation and industrial production better than a random walk and an OLS VAR?
- **Inputs:** monthly 2005:8–2026:7 ($T = 252$): 100 log IP, 100 log HICP, ROBOR 3M, 100 log EUR/RON (BNR), unemployment, ESI; $p = 12$; model: B1
- **Tasks**
 1. Set $\delta_i = 1$ for all variables except the ESI and compute the GLP mode on the full sample.
 2. Forecast recursively from origins 2015:1–2025:7 the 1- and 12-month changes of log HICP and log IP with RW with drift, OLS VAR(12) and the BVAR (hyperparameters re-optimised each January).
 3. Report the RMSE table and the range of the re-optimised λ .
 4. Interpretation: what does the OLS VAR teach about $k = 73$ regressors and 120 observations?
- **Report:** one set of hyperparameters, a table of 12 RMSEs, one interval, one sentence
- **Notebook:** section B2

B3: factors of the FRED-MD panel on three samples [Solved]

- **Question:** is the factor structure of the US economy stable over time?
- **Inputs:** FRED-MD-format panel, transformed and cleaned of outliers; samples 1960–1984, 1985–2007, 2008–2026 (series with at least 80% of observations)
- **Tasks**
 1. Estimate eight factors by EM principal components on each sample.
 2. Report the Bai–Ng choices (IC_{p2} , PC_{p2}), the share of the first component and of the first three.
 3. Report the absolute correlation of the first factor with IP growth, CPI inflation (second difference) and the change of the funds rate.
 4. Interpretation: what changed across the three periods?
- **Report:** a table of three rows and seven columns, one sentence
- **Notebook:** section B3

B3: solution [Solved]



Sample	<i>N</i>	IC_{p2}	PC_{p2}	1st, %	3, %	IP	CPI	FFR
1960–1984	103	6	10	19.4	33.5	0.85	0.10	0.40
1985–2007	109	7	10	13.6	29.2	0.77	0.01	0.54
2008–2026	117	8	10	20.0	42.7	0.80	0.29	0.42

- Interpretation: the first factor is real activity in all periods
 - ▶ it explained least in the Great Moderation and most after 2008 (financial crisis, 2020)
 - ▶ the criteria choose six to eight factors, more in the recent sample

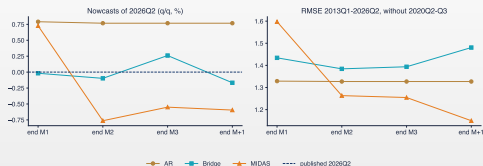
B4: FAVAR and the number of factors [Proposed]

- **Question:** do the FAVAR responses to a monetary policy shock depend on the number of factors K ?
- **Inputs:** balanced FRED-MD-format panel 1960:1–2001:8, two-step FAVAR of [Bernanke, Boivin and Elias \(2005\)](#), VAR(13), funds rate last, 25 bp shock; model: B3
- **Tasks**
 1. Estimate the FAVAR with $K = 1, 3$ and 5 .
 2. Report the IP trough and its month, the largest CPI response within 24 months, the CPI response after 48 months and the unemployment peak.
 3. Interpretation: which K would you report, and how would you defend it before seeing the responses?
- **Report:** a table of three rows and five columns, two sentences
- **Notebook:** section B4

B5: nowcasting Romanian GDP in 2026Q2 [Solved]

- **Question:** how did the AR, bridge and MIDAS nowcasts of 2026Q2 evolve as the data arrived, and how accurate are they in general?
- **Inputs:** monthly panel (Eurostat) with stylised release lags (surveys 0, unemployment 1, hard data 2 months; GDP 2 months after the quarter); published growth 0.00%
- **Tasks**
 1. Build the information sets at the ends of April, May, June and July 2026.
 2. Compute the AR(1), bridge (IP, retail, ESI) and MIDAS (IP, ESI, $K = 6$) nowcasts at each date.
 3. Compute the RMSE of the same models for 2013Q1–2026Q2 without 2020Q2–Q3.
 4. Interpretation: is one quarter enough to rank models?
- **Report:** a table of twelve nowcasts, a table of twelve RMSEs, one sentence
- **Notebook:** section B5

B5: solution [Solved]



End of	AR	Bridge	MIDAS	RMSE AR	RMSE bridge	RMSE MIDAS
April	0.79	−0.02	0.73	1.33	1.43	1.60
May	0.77	−0.10	−0.76	1.33	1.38	1.26
June	0.77	0.26	−0.55	1.33	1.39	1.25
July	0.77	−0.17	−0.60	1.33	1.48	1.15

- Interpretation: one quarter says almost nothing about a ranking
 - ▶ 2026Q2: the bridge nowcast was closest to the published 0.00%, MIDAS overshot downwards
 - ▶ over 2013–2026, MIDAS is best from month 2 on and the bridge worst

B6: the 2026Q3 nowcast and its news [Proposed]

- **Question:** how did the September 2026 releases change the dynamic-factor nowcast of 2026Q3?
- **Inputs:** the panel of B5; two-factor DFM (two-step, Mariano–Murasawa link); information sets at the ends of August and September 2026; model: B5
- **Tasks**
 1. Estimate the DFM on the August information set and compute the nowcast of 2026Q3.
 2. Keeping the parameters fixed, compute the September nowcast and the news decomposition by series.
 3. Check that the impacts add up to the revision; compare with the EM estimate of statsmodels.
 4. Interpretation: which releases moved the nowcast, hard or soft data?
- **Report:** two nowcasts, a table of impacts, one check, two EM numbers, one sentence
- **Notebook:** section B6

C1: do surveys still help after 2020? [Proposed]

- **Question:** do the six Romanian confidence balances improve the end-of-quarter DFM nowcast beyond the hard data, before and after 2020?
- **Inputs:** the panel of B5; DFM with all series (two factors) and with hard data only (one factor); quarters 2013–2019 and 2021–2026; model: B5
- **Tasks**
 1. Write a five-line pre-registration: target, models, information set, sample split, test.
 2. Compute the RMSE of both DFMs and of the AR at the end of month 3 on each subsample.
 3. Interpretation: what can and cannot be concluded from about 25 quarters per subsample?
- **Report:** a five-line plan, a table of six RMSEs, a project plan
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise Bayesian VARs, factor models and nowcasting. The answer:
- ▣ (a) With the Minnesota prior, $\lambda = 0$ gives the OLS estimates.
- ▣ (b) The natural conjugate prior allows a cross-variable shrinkage θ below 1.
- ▣ (c) In large BVARs the optimal λ increases with the number of variables.
- ▣ (d) The sum-of-coefficients prior pushes the VAR towards a model in first differences.
- ▣ (e) Bai-Ng IC_{p2} picks the number of factors that maximises the explained variance.
- ▣ (f) The news of a release is its value; its impact is the value times the loading.
- ▣ (g) A Gibbs sampler with $R\text{-hat} = 1.56$ has converged.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ A prior is information: precisions add, and the marginal likelihood tells how much to trust it
- ▣ Dummy observations turn a Bayesian VAR into OLS on augmented data
- ▣ Principal components and Bai–Ng summarise a large panel; the number of factors can change conclusions
- ▣ Nowcasts improve as data arrive; revisions decompose into news
- ▣ An AI answer is a draft: check limits, counts, signs of diagnostics and definitions

After the seminar

- Lecture 5 develops each topic of today: Minnesota and NIW priors, GLP, large BVARs, factors, FAVAR, DFM, nowcasting of Romanian GDP
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: a real-time nowcasting model for Romania
- Reading: Bańbura, Giannone and Reichlin (2010); Giannone, Lenza and Primiceri (2015); Stock and Watson (2002b); Bańbura and Modugno (2014)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References (1/2)

- Bańbura, M., Giannone, D., & Reichlin, L. (2010). Large Bayesian vector auto regressions. *Journal of Applied Econometrics*, 25(1), 71–92. doi:10.1002/jae.1137
- Bańbura, M., & Modugno, M. (2014). Maximum likelihood estimation of factor models on datasets with arbitrary pattern of missing data. *Journal of Applied Econometrics*, 29(1), 133–160. doi:10.1002/jae.2306
- Baffigi, A., Golinelli, R., & Parigi, G. (2004). Bridge models to forecast the euro area GDP. *International Journal of Forecasting*, 20(3), 447–460. doi:10.1016/S0169-2070(03)00067-0
- Bai, J., & Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1), 191–221. doi:10.1111/1468-0262.00273
- Bernanke, B. S., Boivin, J., & Elias, P. (2005). Measuring the effects of monetary policy: A factor-augmented vector autoregressive (FAVAR) approach. *The Quarterly Journal of Economics*, 120(1), 387–422. doi:10.1162/0033553053327452
- Doan, T., Litterman, R., & Sims, C. (1984). Forecasting and conditional projection using realistic prior distributions. *Econometric Reviews*, 3(1), 1–100. doi:10.1080/07474938408800053
- Gelman, A., & Rubin, D. B. (1992). Inference from iterative simulation using multiple sequences. *Statistical Science*, 7(4), 457–472. doi:10.1214/ss/1177011136
- Ghysels, E., Santa-Clara, P., & Valkanov, R. (2006). Predicting volatility: Getting the most out of return data sampled at different frequencies. *Journal of Econometrics*, 131(1–2), 59–95. doi:10.1016/j.jeconom.2005.01.004
- Giannone, D., Lenza, M., & Primiceri, G. E. (2015). Prior selection for vector autoregressions. *The Review of Economics and Statistics*, 97(2), 436–451. doi:10.1162/REST_a_00483
- Kadiyala, K. R., & Karlsson, S. (1997). Numerical methods for estimation and inference in Bayesian VAR-models. *Journal of Applied Econometrics*, 12(2), 99–132. doi:10.1002/(SICI)1099-1255(199703)12:2<99::AID-JAE429>3.0.CO;2-A
- Litterman, R. B. (1986). Forecasting with Bayesian vector autoregressions: Five years of experience. *Journal of Business & Economic Statistics*, 4(1), 25–38. doi:10.1080/07350015.1986.10509491
- Mariano, R. S., & Murasawa, Y. (2003). A new coincident index of business cycles based on monthly and quarterly series. *Journal of Applied Econometrics*, 18(4), 427–443. doi:10.1002/jae.695

References (2/2)

- Sims, C. A., & Zha, T. (1998). Bayesian methods for dynamic multivariate models. *International Economic Review*, 39(4), 949–968. doi:[10.2307/2527347](https://doi.org/10.2307/2527347)
- Stock, J. H., & Watson, M. W. (2002a). Forecasting using principal components from a large number of predictors. *Journal of the American Statistical Association*, 97(460), 1167–1179. doi:[10.1198/016214502388618960](https://doi.org/10.1198/016214502388618960)
- Stock, J. H., & Watson, M. W. (2002b). Macroeconomic forecasting using diffusion indexes. *Journal of Business & Economic Statistics*, 20(2), 147–162. doi:[10.1198/073500102317351921](https://doi.org/10.1198/073500102317351921)